StudyDock AI

An AI Tutor Program Based on Documents

Using HTML

A PROJECT REPORT

Submitted by

Roshan

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ABSTRACT

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This project, called **StudyDock AI**, is a web-based intelligent learning assistant that helps students study more effectively using their own PDF documents. It uses an external AI model (Gemini API) to understand study material and generate explanations or quizzes based on user input and uploaded content.

The system works like this:

1. It allows users to upload PDF documents and extracts all text content from them so it can be used as study context.
2. It processes the extracted data and uses it as input context for an AI model to generate accurate responses based on the uploaded study material.
3. It provides a user-friendly interface with two modes:
    - **Tutor Mode**, where users can ask questions and get explanations
    - **Quiz Mode**, where the system generates quizzes for revision
4. It manages errors such as missing API keys or invalid inputs and ensures smooth interaction through a simple web-based interface.

StudyDock AI makes it easier for students to interact with their study materials instead of only reading them passively. It combines document processing and artificial intelligence to create a smarter and more interactive learning experience.

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Chapter 1
Introduction

StudyDock AI is an AI-powered learning assistant built using web technologies. It is designed to help students understand study materials more effectively by combining a PDF viewer with an intelligent chatbot system. The application uses the Gemini API to analyze uploaded documents and provide explanations, answers, and quizzes based on the content.

By integrating a clean and responsive web interface, StudyDock AI offers a simple and accessible learning environment where students can interact with their study material instead of only reading it passively.

1.1 Objective

The primary objective of this project is to design and implement an AI-based learning assistant that improves the way students interact with study materials. The specific objectives include:

1. Enhance Accessibility to Learning Materials
    - Provide a platform where students can upload PDFs and directly interact with their content using AI assistance.
2. Utilize Artificial Intelligence for Learning Support
    - Use the Gemini AI model to analyze study material and generate explanations, answers, and quizzes based on user queries.
3. Simplify Study and Revision Process
    - Convert static PDF content into an interactive learning experience with tutoring and quiz generation features.
4. Provide Dual Learning Modes
    - Offer both Tutor Mode (for explanations and doubt solving) and Quiz Mode (for self-assessment and revision).

The overall goal is to improve learning efficiency by combining document reading and AI assistance into a single platform, helping students understand concepts more effectively.

1.2 Problem Statement

Students often face difficulties in efficiently understanding large volumes of study material from static PDF files or textbooks. Traditional learning methods can be time-consuming and lack interactivity.

The key challenges include:

1. Time-Consuming Manual Learning
   - Students spend significant time searching and understanding concepts within large documents.
2. Lack of Interactive Study Tools
   - Most PDF readers do not provide explanations or learning assistance.
3. Difficulty in Self-Assessment
   - Students often lack quick ways to test their understanding of study material.
4. Fragmented Learning Experience
   - Switching between notes, search engines, and study tools reduces focus and efficiency.

This project addresses these issues by providing an **AI-powered** system that allows students to directly interact with their study material, receive explanations, and generate quizzes, improving both understanding and revision efficiency.

1.3 Project Overview

StudyDock AI is a web-based intelligent learning assistant designed to help students study more effectively using uploaded PDF documents. It integrates a document viewer with an AI chatbot that can analyze content and respond to user queries.

The system provides two main modes:

- Tutor Mode: Helps explain concepts and answer questions based on uploaded material
- Quiz Mode: Generates quizzes to test understanding and support revision

By combining document processing and AI interaction, StudyDock AI creates a unified learning environment that improves engagement and understanding.

1.4 Introduction to Artificial Intelligence

Artificial Intelligence (AI) refers to the ability of machines or software systems to perform tasks that typically require human intelligence. This includes understanding language, solving problems, and making decisions.

AI systems are widely used in various fields such as education, healthcare, automation, and recommendation systems.

In StudyDock AI, artificial intelligence is used to:

- Understand and process user queries
- Analyze uploaded PDF content
- Generate explanations in Tutor Mode
- Create quizzes in Quiz Mode

The application relies on a large language model (Gemini API), which processes natural language input and generates meaningful responses based on context.

Chapter 2

System Description

StudyDock AI is a web-based application developed using HTML, CSS, and JavaScript. It integrates a PDF viewer with an AI-powered chatbot system. The system allows users to upload study materials in PDF format and interact with them using artificial intelligence.

Instead of using traditional machine learning models trained on datasets, StudyDock AI uses the Gemini API to process user queries and generate responses dynamically based on the uploaded document content.

2.1 HTML

HTML (HyperText Markup Language) is the standard markup language used to structure web pages. In StudyDock AI, HTML is used to define the layout of the application, including the document panel, PDF viewer, and AI chat interface.

It provides the basic structure for:

- Sidebar for uploaded documents
- Central PDF viewer
- Right-side AI chat system

2.2 Tailwind CSS

Tailwind CSS is a utility-first CSS framework used for styling the application. It allows rapid UI development by using predefined classes instead of writing custom CSS.
In StudyDock AI, Tailwind CSS is used to:

- Create a responsive layout
- Support dark mode UI
- Style buttons, inputs, and chat components
- Ensure a clean and modern user interface

2.3 JavaScript

JavaScript is used to add interactivity and functionality to the application. It handles the logic behind file uploads, chat interaction, and communication with the Gemini API.
Key functions of JavaScript in this project include:

- Managing application state (chat, documents, modes)
- Handling user input and chat messages
- Sending requests to the Gemini API
- Updating the UI dynamically without page reload.

2.4 PDF.js

PDF.js is a JavaScript library used to display and extract content from PDF files directly in the browser.

In StudyDock AI, PDF.js is used to:

- Load and render PDF documents
- Extract text content from uploaded files
- Enable page navigation (next/previous)
- Support zoom functionality

This allows users to view and analyze their study materials without leaving the application.

2.5 Gemini API

The Gemini API is an artificial intelligence model provided by Google that processes natural language input and generates intelligent responses.

In StudyDock AI, it is used to:

- Answer user questions based on uploaded documents
- Explain concepts in simple language (Tutor Mode)
- Generate quizzes for revision (Quiz Mode)
- Provide context-aware responses using extracted PDF text

Unlike traditional machine learning models, the Gemini API does not require training on local datasets. Instead, it processes input dynamically in real time.

Chapter 3
System Design

3.1 Document Upload and Processing

The first major component of StudyDock AI is the **document upload and processing system**. This module allows users to upload one or more PDF files that contain their study material. Instead of relying on a fixed built-in dataset, the system uses the uploaded documents as the primary source of learning content.

When a user selects PDF files, the application processes them using **PDF.js**, a JavaScript library for rendering and reading PDF documents in the browser. Each PDF is loaded page by page, and the text content is extracted from every page. The extracted text is then combined into a single text block for that document and stored in the application state along with other document information such as file name, current page number, zoom level, and the PDF object itself.

This design allows the application to support multiple uploaded study documents at the same time. The user can switch between uploaded files from the document panel, preview the pages in the built-in PDF viewer, and use the combined extracted text as learning context for AI-based tutoring and quiz generation. By directly processing the user's own study material, StudyDock AI becomes flexible enough to work with a wide range of academic subjects.

3.2 AI Model Integration

StudyDock AI integrates with the **Google Gemini API** to provide intelligent tutoring and quiz generation features. Instead of using a separately trained machine learning model, the system sends the uploaded study material and the user's query to the **Gemini** model, which then generates a contextual response. This allows the application to answer questions and create quizzes based on the actual content of the uploaded **PDFs**.

Whenever a user sends a message, the system builds a prompt containing the extracted PDF text, optional student details such as name and age, and the user's input. This prompt is sent to the **Gemini 2.5 Flash** model through an **API request**, and the generated response is displayed in the chat panel. If an error occurs during the request, the system shows an appropriate error message to the user.

3.3 Tutor Mode Logic

Tutor Mode is one of the two core interaction modes in StudyDock AI. It is designed to act as an AI study assistant that explains concepts, answers questions, and simplifies difficult material from the uploaded PDFs.

When the user enters a question in Tutor Mode, the system first determines the currently active mode and locks it so that the response is correctly added to the Tutor chat history. It then gathers the extracted text from all uploaded documents and combines it into a single context block. If the total text becomes too large, the system trims it to a manageable length before sending it to the AI model.

A Tutor Mode prompt is then generated in the following structure:

1. **Optional** student profile details.
2. The uploaded study material as context.
3. A Question asked to the AI to explain the topic in a way that makes studying easier.

The **response** generated by the **Gemini API** is displayed in the Tutor chat area. Tutor Mode is intended to help students understand difficult concepts, revise topics, and clarify doubts using their own uploaded notes or textbook material instead of generic internet content.

3.4 Quiz Generation Logic

Quiz Generation Logic in StudyDock AI is responsible for creating quizzes based on the uploaded study material. When the user enters a request in Quiz Mode, the system gathers the extracted text from the uploaded PDF documents along with optional student details such as name and age, and combines them into a prompt for the Gemini API. The prompt asks the AI to generate quiz content based on the selected topic or study material provided by the user. The generated response is then displayed in the Quiz chat area, allowing students to test their understanding of the uploaded content in an interactive way. By keeping quiz conversations separate from Tutor Mode conversations, the application ensures a cleaner and more organized learning experience.

Chapter 4

Program Functionality and Workflow

The StudyDock AI system is designed to function as an AI-powered learning assistant that provides tutoring and quiz generation based on user-uploaded study material. The system consists of two main components: the **Backend AI Integration** and the **Frontend User Interface (UI)**.

4.1 Backend Model

The backend of StudyDock AI is built around the Google Gemini API, which handles both tutoring responses and quiz generation. Instead of using a traditional machine learning model, the system relies on a generative AI model that processes user input along with extracted text from uploaded PDF documents.

When a user sends a query, the system collects the chat input, student profile data, and the extracted document content, then combines them into a structured prompt. This prompt is sent to the Gemini API, which generates a contextual response based on the provided study material. The backend also manages separate chat histories for Tutor Mode and Quiz Mode to ensure organized interaction and proper context handling.

4.2 Frontend User Interface (UI)

The User Interface of StudyDock AI is built using HTML, Tailwind CSS, and JavaScript, providing a clean and responsive design. It is divided into three main sections: document upload panel, PDF viewer, and AI chat system.

Users can upload multiple PDF files, view them inside an integrated PDF viewer, and interact with the AI through Tutor Mode and Quiz Mode tabs. The chat interface allows users to send questions and receive AI-generated responses in real time. Additional features include API key input, name and age fields for personalization, and theme switching for light/dark mode support. The UI is designed to be simple, interactive, and student-friendly.

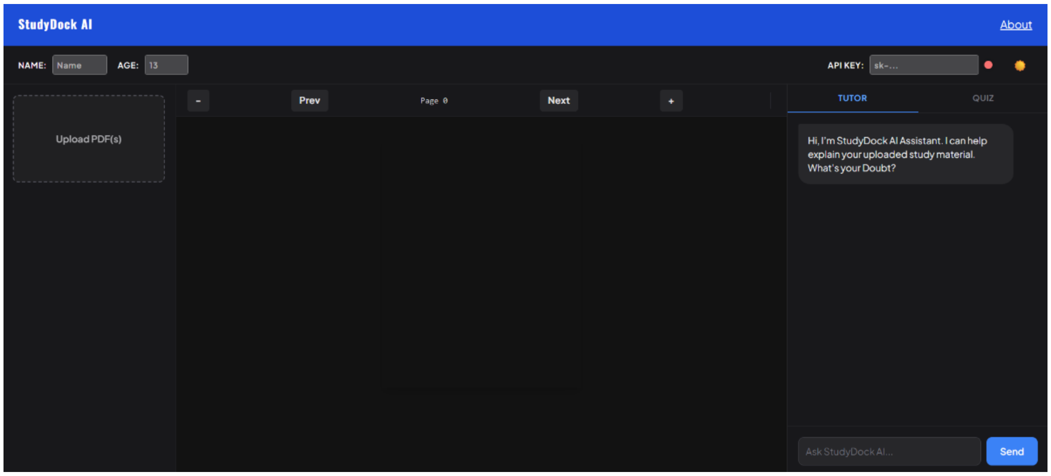

Figure 4.1 Program Screenshot

4.3 Dataset

Instead of using a fixed dataset, StudyDock AI dynamically uses **user-uploaded PDF** documents as its **primary data source**. These documents are processed using PDF.js, which extracts text content from each page of the uploaded files.

The extracted text is stored in memory and used as the knowledge base for AI responses. This approach allows the system to work with any subject or study material provided by the user, making it flexible and adaptable rather than limited to a predefined dataset.

4.4 System Workflow

1. **Document Input:** The user uploads one or more PDF files, which are processed and converted into readable text.
2. **Context Preparation:** When the user asks a question or requests a quiz, the system combines the extracted document text with user input and optional profile details.
3. **AI Processing:** The prepared prompt is sent to the Gemini API, which generates a response based on the study material.
4. **Result Display:** The AI response is shown in the chat interface under either Tutor Mode or Quiz Mode, depending on the selected tab.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

StudyDock AI successfully combines artificial intelligence with an interactive and user-friendly interface to create a useful study assistant for students. By integrating PDF text extraction, AI-powered tutoring, and quiz generation into a single platform, the system helps users learn from their study material in a more engaging and personalized way. The application allows students to upload documents, ask questions, receive explanations, and test their understanding through quizzes, all within the same environment.

This project demonstrates how AI can be applied in education to simplify studying and improve access to learning support. The combination of document processing, chat-based interaction, and a clean interface makes StudyDock AI a practical and flexible tool for academic use. Overall, the project highlights the potential of combining modern web technologies with generative AI to create smarter educational applications.

5.2 Future Work

StudyDock AI has strong potential for further development and improvement. Some possible future enhancements include:

- **Improved Quiz Features:** Add support for multiple-choice questions, scoring, answer checking, and progress tracking to make Quiz Mode more interactive.
- **Better PDF Support:** Improve document processing for scanned PDFs, images, and more complex layouts to make the system work with a wider range of study material.
- **Mobile App Enhancements:** Expand the mobile app version with a more optimized layout and features designed specifically for smaller screens.
- **Smarter Study Tools:** Add features such as flashcard generation, note summarization, topic-wise revision, and study progress tracking to make the platform more comprehensive.

APPENDIX

Index.html

The Full code can be viewed by Clicking Here